Script-Controlled ACL: Restrict Record Access Based on Field Value

Project Design Phase-II: Solution Requirements (Functional & Non-functional)

Date: October 3, 2026	**Maximum Marks:** 4 Marks
Team ID: SWTID-2026-8089	**Project Name:** Script-Controlled ACL

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User & Role Setup	Administrator can create the EEE test user under User Administration → Users (USN-1) Administrator can create the custom roles bb1, bb2, bb3 and bb4 under User Administration → Roles and assign all four roles to the EEE User (USN-1)
FR-2	Custom Table Creation	Create the Institution Details table (u_institution_details) under System Definition → Tables with the fields Student Roll Number (auto number), Student Name, Faculty Name, Branch (choice: ECE, EEE, CSE), Email, Phone Number and Description (USN-1) Add multiple records with different Branch values (ECE, EEE, CSE) so that the access rules can be tested (USN-1)
FR-3	Read Access Control (READ ACL)	Elevate the security_admin role and create a READ ACL (Type: record, Operation: read) on u_institution_details with Advanced selected (USN-2) Add the role bb1 in the Requires role related list, set the data condition Branch is EEE, and add the script that allows admin and bb1 users and denies all others (USN-2) A user with bb1 sees only EEE records, a user without the role sees no records, and the admin sees all records (USN-2)
FR-4	Create Access Control (CREATE ACL)	Create a CREATE ACL (Operation: create) on u_institution_details with the role bb2 and no data condition (USN-3) A user with the bb1 and bb2 roles can see the EEE records along with the New button (USN-3)
FR-5	Write Access Control (WRITE ACL)	Create a WRITE ACL (Operation: write) on u_institution_details with the role bb3 and no data condition (USN-4) A user with the bb1, bb2 and bb3 roles can edit fields such as Description in the EEE records (USN-4)
FR-6	Delete Access Control (DELETE ACL)	Create a DELETE ACL (Operation: delete) on u_institution_details with the role bb4 and no data condition (USN-5) A user with the bb1, bb2, bb3 and bb4 roles can select an EEE record and delete it using Actions on selected rows (USN-5)

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-7	Validation & Testing	Impersonate the EEE user, a user without roles and the Admin, and open the Institution Details list (u_institution_details.list) for each (USN-2 to USN-5) Confirm that each user sees and can do only what the assigned roles allow: view, create, edit and delete for EEE records, no records for a user without roles, and all records for the admin (USN-2 to USN-5)

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	Administrators create the rules through a simple path (System Security → Access Control (ACL) → New) using clearly labelled fields. Users open the Institution Details list as usual and see the New button, inline edit and Delete action only when they have the matching role (bb2, bb3, bb4).
NFR-2	Security	Access is controlled by ServiceNow login and role-based ACLs. The READ ACL checks both the user's role (bb1) and the record's Branch value (EEE), so users without the role see no records. Administrators keep full access. Only a user who has elevated to the security_admin role can create or change ACLs.
NFR-3	Reliability	The ACLs are evaluated automatically every time a user opens a list or form, so no record is shown without being checked. The script returns true only for allowed users and returns false for everyone else, so access is denied by default.
NFR-4	Performance	The ACL check runs on the server when records are requested and uses only a simple role check and one data condition. Users see the filtered Institution Details list without noticeable delay.
NFR-5	Availability	The solution runs on the ServiceNow cloud platform, so administrators and users can use it from a web browser whenever the ServiceNow instance is available.
NFR-6	Scalability	The same ACLs apply to any number of records and users, because access depends on roles and field values. The same approach can be extended to other branches (ECE, CSE) and to other tables by adding roles, data conditions or new ACLs.